

EXPLORATORY DATA ANALYSIS & INSIGHTS USING SHOPIFY STOCK DATASET

OBJECTIVE:

The Shopify Stock Dataset was analysed using Python to understand stock price movements, daily returns, trading volume, and unusual observations. Data cleaning, statistical analysis, and visualization techniques were used to identify important patterns in the dataset.

ANALYSIS AND KEY FINDINGS:

Analysis	Key Finding	Insight
Data Cleaning	No missing values or duplicate records were found.	The dataset was suitable for further analysis
Daily Price Change	Daily price change was calculated using Close – Open.	It shows the daily movement in stock price.
Daily Return	Daily return was calculated as a percentage.	It helps understand the daily percentage change in price.
Trading Volume	Volume showed considerable variation across the dataset.	High-volume days can indicate unusual trading activity.
Anomaly Detection	IQR method was used to identify high-volume observations.	Unusual volume observations can be investigated further.
Return Distribution	A histogram was used to visualize Daily Return.	It helps understand the distribution of daily price movements.

MAJOR INSIGHTS:

- The dataset was successfully cleaned and prepared for analysis.
- Daily Price Change and Daily Return provide useful information about stock movement.
- Trading volume varies across different dates.
- Some observations have unusually high trading volume.
- Visualization helps understand the distribution and patterns in the stock data.

RECOMMENDATIONS:

- Monitor high-volume trading days for further analysis.
- Compare trading volume with daily returns to identify possible relationships.
- Use time-series visualizations to study stock price trends.
- Further statistical analysis can be performed to understand price and volume relationships.

CONCLUSION:

The EDA of the Shopify Stock Dataset helped in understanding stock price movements, daily returns, trading volume, and unusual observations. Data cleaning, statistical methods, and visualization provided useful insights and prepared the dataset for further analysis.